Predicting probability of scoring a second date

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1 Problem Description

We will be predicting whether participants of a first date will go on a second date from features such as attractiveness, sincerity, intelligence, and ambition. Given the property of our dataset, we will construct a Bayesian Network with hidden variables to model the problem, learn CPTs using Expectation Maximization, and apply exact inference for prediction. We will be analyzing how the raw features affect the hidden variables and how the hidden variables influence the outcome of a second date.

2 Dataset Source

We will be using the Kaggle *Speed Dating* dataset. It contains information from 8,378 participants from the years 2002 to 2004.

Each row represents one participant and their date. We plan to use 3 attributes from the dataset, and for each attribute (attractiveness, intelligence, sincerity), the dataset provides 5 related features.

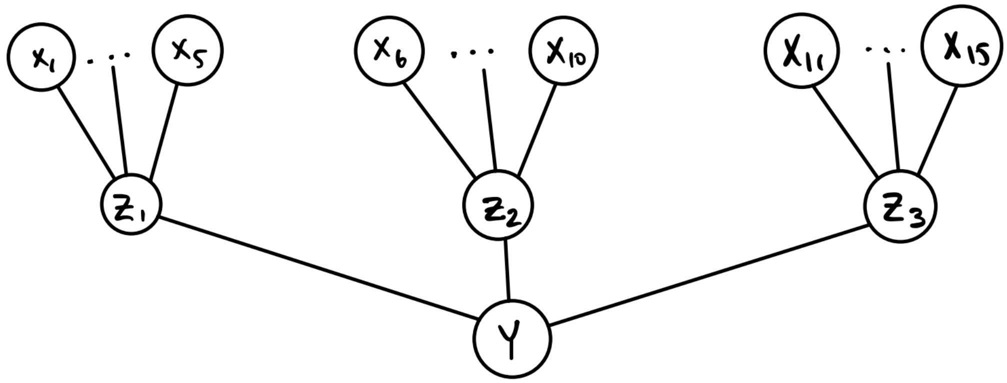
For example, for **attractiveness**, the dataset includes:

- `attractive`: Participant's self-rating on attractiveness
- `attractive_partner`: Participant's rating of their partner's attractiveness
- `attractive_o`: Partner's rating of the participant's attractiveness during the event
- `attractive_important`: How important attractiveness is to the participant when choosing a partner
- `pref_o_attractive`: How important the participant believes their partner considers attractiveness

We will include $5 \times 3 = 15$ features in total for our task.

3 Methodology

Given raw features $\vec{X}$, we want to infer the probability of a second date Y. We want to build a model using the dataset $V_t = \{\vec{X}, Y\}$. We constructed 3 hidden variables $\vec{Z} = \{Z_1, Z_2, Z_3\}$, each Z represents an attribute (attractiveness, intelligence, sincerity) and incorporates information from 5 features. We want to find out how the raw features $\vec{X}$ affect the hidden variables $\vec{Z}$ and how the hidden variables $\vec{Z}$ influence the outcome of a second date.



Let T be the cardinality of the dataset, x_i be the features, z_i be the hidden variables, and $Y \in \{0, 1\}$ be whether a second date occurs.

We want to maximize the likelihood $P(\vec{X}, Y)$ using EM.

E-step:

$$P(\vec{Z}|\vec{X}, Y) = \frac{P(\vec{X}, \vec{Z}, Y)}{P(\vec{X}, Y)} = \frac{P(\vec{X})P(\vec{Z}|\vec{X})P(Y|\vec{Z})}{\sum_{\vec{Z}'} P(\vec{X}, \vec{Z}', Y)} = \frac{P(\vec{X})P(\vec{Z}|\vec{X})P(Y|\vec{Z})}{\sum_{\vec{Z}'} P(\vec{X})P(\vec{Z}'|\vec{X})P(Y|\vec{Z}')}$$
$$= \frac{P(\vec{Z}|\vec{X})P(Y|\vec{Z})}{\sum_{\vec{Z}'} P(\vec{Z}'|\vec{X})P(Y|\vec{Z}')}$$

M-step:

Root nodes:

$$P(\vec{X}) \leftarrow \frac{\sum_{i=1}^{T} P(\vec{X}|\vec{X}_i, Y_i)}{T} = \frac{\sum_{i=1}^{T} I(\vec{X}, \vec{X}_i)}{T}$$

Nodes with parents:

$$P(Z|\vec{X}) \leftarrow \frac{\sum_{i=1}^{T} P(\vec{X}, Z|\vec{X}_i, Y_i)}{\sum_{i=1}^{T} P(\vec{X}|\vec{X}_i, Y_i)} = \frac{\sum_{i=1}^{T} I(\vec{X}, \vec{X}_i)P(\vec{Z}|\vec{X}_i, Y_i)}{\sum_{i=1}^{T} I(\vec{X}, \vec{X}_i)}$$

$$P(Y|\vec{Z}) \leftarrow \frac{\sum_{i=1}^{T} P(Y, \vec{Z}|\vec{X}_i, Y_i)}{\sum_{i=1}^{T} P(Z|\vec{X}_i, Y_i)} = \frac{\sum_{i=1}^{T} I(Y, Y_i)P(\vec{Z}|\vec{X}_i, Y_i)}{\sum_{i=1}^{T} P(Z|\vec{X}_i, Y_i)}$$